

Mei-Yu Lai

Computational Biologist

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SUMMARY

Computational biologist with 8+ years of experience analyzing large-scale human genetics and multi-omics datasets to identify, prioritize, and validate therapeutic targets. Hands-on experience with GWAS, pQTL, colocalization, fine-mapping, and variant prioritization across population-scale and rare disease cohorts, using bioinformatics tools (BCFtools, BEDtools, PLINK) on HPC infrastructure. Deep expertise in functional genomics, single-cell transcriptomics, proteogenomics, and statistical genetics. Proven ability to lead cross-functional research and translate complex analytical insights into actionable biological discoveries.

CORE SKILLS

Human Genetics & Statistical Genetics: GWAS, pQTL, colocalization, fine-mapping, Mendelian randomization, causal variant prioritization

Bioinformatics Tools: BCFtools, BEDtools, PLINK, VCF and BED/BAM formats, variant annotation and curation, STAR

Programming & Computing: Python, R, MATLAB, Shell scripting, Unix/Linux, HPC/Slurm, Git

Multi-Omics & Functional Genomics: Multi-omics integration (bulk RNA-seq, proteomics, metabolomics), single-cell RNA-seq, biomarker discovery, predictive modeling, survival and longitudinal analysis

Communication: Technical presentations at international conferences; cross-functional collaboration with wet-lab and clinical teams

EXPERIENCE

Data Scientist

04/2022 – Present

Beth Israel Deaconess Medical Center, Boston, MA

- Built end-to-end proteogenomic pipelines on HPC infrastructure integrating large-scale GWAS and pQTL datasets to identify and prioritize putative causal proteins
- Processed and curated variant-level data using BCFtools and BEDtools for variant filtering and file format conversion
- Implemented single-cell transcriptomic analyses that resolved cell-type-specific gene expression patterns underlying meningeal lymphatic dysfunction in PD
- Developed multi-omics analytical frameworks to identify novel biomarkers and built machine learning models for disease subtype stratification

Bioinformatician

06/2021 – 09/2021

Johns Hopkins University, Baltimore, MD

- Prioritized causal variants in rare disease in Critical Assessment of Genome Interpretation (CAGI6) challenge
- Tested and evaluated causal variant predictions and interpreted their clinical relevance

Biostatistician

08/2012 – 06/2016

Academia Sinica, Taiwan

- Investigated differentially expressed genes in breast cancer and developed statistical models for prognosis of distant metastasis
- Designed and constructed an in-house web tool for sample size calculations across multiple study designs
- Analyzed clinical data to support the development of clinical trial protocols

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PROJECTS

Proteogenomic Insights of Parkinson's Disease

Discovered putative causal PD proteins using a systems biology approach that could serve as therapeutic targets while providing a novel framework for understanding PD pathogenesis and disease prediction.

Multi-Omics Approaches Applied to Neurodegenerative Diseases

Used high-throughput omics technologies to uncover potential biomarkers to improve our understanding of underlying disease mechanisms, predict disease causality, and improve the design of clinical trials.

Cellular Atlas of Parkinson's Disease

Uncovered cell-type-specific differentially expressed genes to decipher the role of meningeal lymphatics in PD.

Gut-Brain Axis in Parkinson's Disease

Identified signatures to distinguish PD patients at higher risk of developing severe gastrointestinal disease using a multi-omic approach, thereby benefiting personalized medicine.

BRCA1 Gene Expression Driven Metastatic Risk in ER+ Breast Cancer

Employed a logistic-AFT mixture regression model to elucidate the link between BRCA1 gene expression and aggressive tumor behavior.

Longitudinal Data Analysis for Metabolic Profiles in Polycystic Ovary Syndrome

Utilized generalized estimating equations (GEE) within a segmented regression model to assess the heterogeneity in response to treatment duration.

EDUCATION

MSE, Biomedical Engineering (Genomics & Systems Biology)

12/2021

Johns Hopkins University, Baltimore, MD

MS, Biometry

06/2012

National Taiwan University, Taiwan

BS, Applied Mathematics

06/2010

Chung Yuan Christian University, Taiwan

SELECTED PRESENTATIONS

Digestive Disease Week (Talk), Washington DC

05/2024

Lai MY, Kulkarni S, and Benitez BA. Molecular Markers Associated with Self-reported Progressive Gut Dysfunction in Parkinson's Disease Patients.

American Academy of Neurology Annual Meeting (Talk), Boston, MA

04/2023

Lai MY, Palanisamy D, and Benitez BA. Proteomic Signatures Associated with Genetic and Clinical Subtypes of Parkinson's Disease.